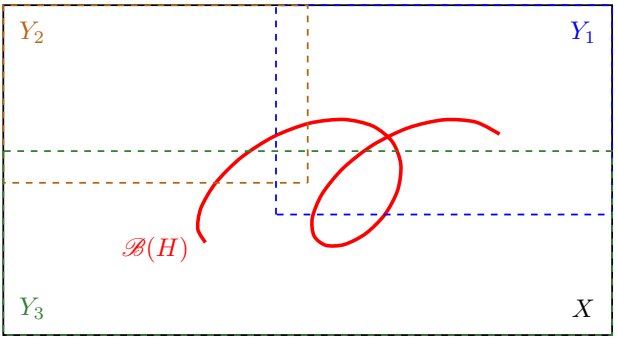
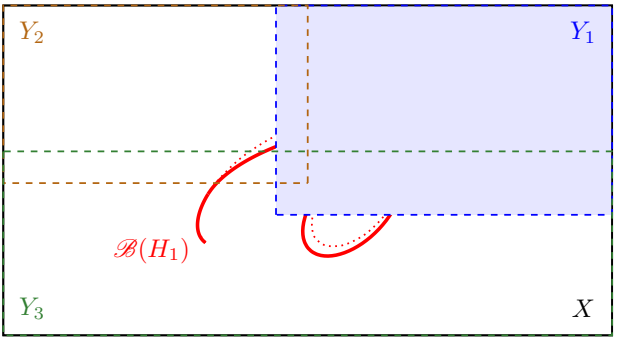
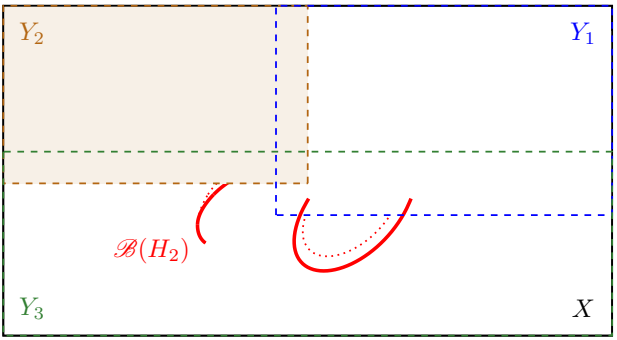


FIGURE 5. The proof that $\mathbb{M}$ is dense goes as follows.

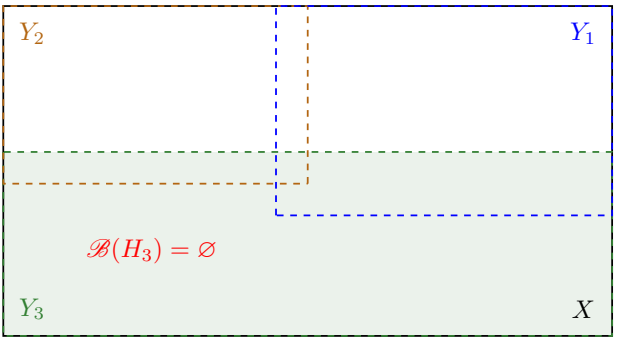
(a) We first cover X by topologically trivial open sets (here Y_1, Y_2, Y_3) on which the degeneracies of H reduce to those of a 2×2 system.



(b) In Y_1, the degenerate part of H reduces to that of a 2×2 system. Via the procedure of §3.1, we can produce H_1, arbitrarily close to H, with no bad points in Y_1. By Lemma 3.3, $\mathscr{B}(H_1)$ is a small perturbation of $\mathscr{B}(H) \smallsetminus Y_1$.



(c) We repeat the procedure and produce H_2, arbitrarily close to H_1, with no bad points in Y_2. As bad points are stable, $\mathscr{B}(H_2)$ is close to $\mathscr{B}(H_1)$. In particular passing from H_1 to H_2 does not generate bad points back in $Y_1 \smallsetminus (Y_2 \cup Y_3)$, and removes bad points in Y_2.



(d) We get new systems H_1, H_2, H_3, recursively constructed, arbitrarily close to H, with no bad points in $Y_1, Y_1 \cup Y_2 \smallsetminus Y_3, Y_1 \cup Y_2 \cup Y_3$, respectively. Since $Y_1 \cup Y_2 \cup Y_3$ cover X, H_3 is in $\mathbb{M}$ and is arbitrarily close to H.